

Lecture 16 - Class Activity MANOVA

Bill Perry

```
# Load required packages
library(car)           # For ANOVA tests
library(emmeans)       # For estimated marginal means
library(mvnormtest)    # For multivariate normality test
library(biotools)      # For Box's M test
library(candisc)       # For canonical discriminant analysis
library(heplots)       # For multivariate plots
# library(MASS)        # For linear discriminant analysis
library(broom)         # For model summaries
library(patchwork)     # For combining plots
library(janitor)       # For cleaning names
library(tidyverse)     # For data manipulation and visualization

# # Set options
# options(scipen = 999)
```

Lecture 16: Multivariate Analysis of Variance (MANOVA)

What is MANOVA?

MANOVA (Multivariate Analysis of Variance) extends ANOVA to multiple response variables: - Compares group centroids in multivariate space - Tests whether groups differ on multiple dependent variables simultaneously - Controls family-wise error rate - Accounts for correlations between dependent variables

When to Use MANOVA

Use MANOVA when you have: - **Response variables:** Multiple continuous variables (correlated) - **Predictor variable:** One or more categorical variables (factors/groups) - **Goal:** Test for group differences across all response variables simultaneously

Key Assumptions of MANOVA

1. **Independence** of observations
2. **Multivariate normality** within groups
3. **Homogeneity of covariance matrices** (Box's M test)
4. **No extreme multivariate outliers**
5. **Linear relationships** among dependent variables
6. **No multicollinearity** (but some correlation is expected)

! Critical First Step

Always check **multivariate normality** and **homogeneity of covariance matrices** before proceeding with MANOVA. These assumptions are more stringent than univariate ANOVA.

Part 1: Iris Data Analysis

Data Overview

We'll analyze morphological measurements of three iris species: - *Iris setosa* - *Iris versicolor*
- *Iris virginica*

We have four measurements: sepal length, sepal width, petal length, and petal width. MANOVA will test whether species differ across all four measurements simultaneously.

```
# Load the iris data from the data subdirectory
iris_df <- read_csv("data/iris.csv")
```

```
Rows: 150 Columns: 5
— Column specification —————
Delimiter: ","
chr (1): species
dbl (4): sepal_length, sepal_width, petal_length, petal_width

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Clean names and prepare data
iris_df <- iris_df %>% clean_names()

# View data structure
head(iris_df)
```

```
# A tibble: 6 × 5
  sepal_length sepal_width petal_length petal_width species
      <dbl>       <dbl>       <dbl>       <dbl> <chr>
1         5.1         3.5         1.4         0.2 setosa
2         4.9         3         1.4         0.2 setosa
3         4.7         3.2         1.3         0.2 setosa
4         4.6         3.1         1.5         0.2 setosa
5          5         3.6         1.4         0.2 setosa
6         5.4         3.9         1.7         0.4 setosa
```

```
# Check sample sizes by species
iris_df %>%
  count(species)
```

```
# A tibble: 3 × 2
  species      n
  <chr>    <int>
1 setosa     50
2 versicolor 50
3 virginica   50
```

```
# Create long format for visualization
iris_long_df <- iris_df %>%
  pivot_longer(
```

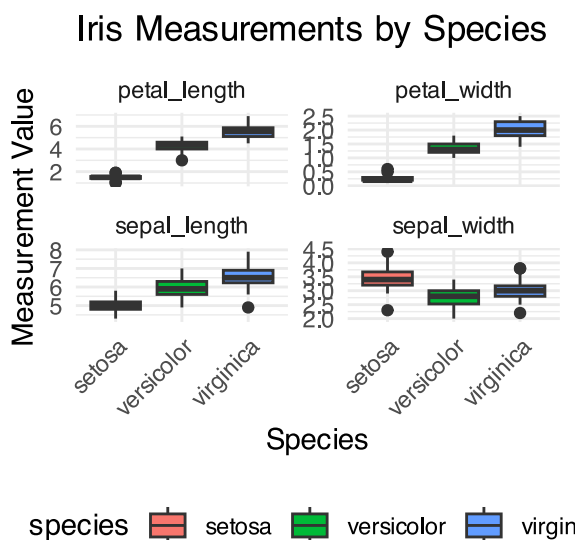
```

    cols = -species,
    names_to = "variable",
    values_to = "measure"
  )

# Plot all variables by species
iris_boxplot <- iris_long_df %>%
  ggplot(aes(species, measure, fill = species)) +
  geom_boxplot() +
  facet_wrap(~ variable, scales = "free_y") +
  theme_minimal() +
  labs(title = "Iris Measurements by Species",
       x = "Species",
       y = "Measurement Value") +
  theme(legend.position = "bottom",
        axis.text.x = element_text(angle = 45, hjust = 1))

iris_boxplot

```



Step 1: Visualize Relationships Between Variables

Before running MANOVA, let's examine the correlations between our response variables.

```

# Calculate means by species to show centroids
mean_points_df <- iris_df %>%
  group_by(species) %>%
  summarise(
    mean_petal_length = mean(petal_length),
    mean_petal_width = mean(petal_width),
    .groups = 'drop'
  )

# Create scatterplot with centroids
iris_centroid_plot <- iris_df %>%
  ggplot(aes(x = petal_length, y = petal_width, color = species)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_point(data = mean_points_df,
            aes(x = mean_petal_length, y = mean_petal_width, fill = species),

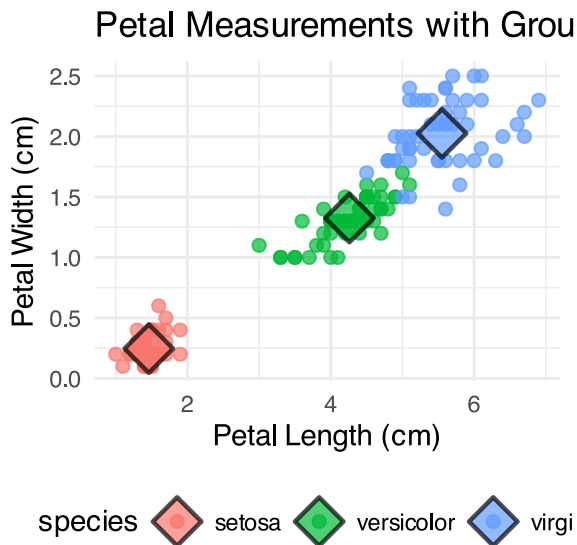
```

```

    shape = 23, color = "black", stroke = 1.2, alpha = 0.7,
    size = 6) +
theme_minimal() +
labs(title = "Petal Measurements with Group Centroids",
     x = "Petal Length (cm)",
     y = "Petal Width (cm)") +
theme(legend.position = "bottom")

```

iris_centroid_plot



Step 2: Test Assumptions

Multivariate Normality

```

# Extract response variables as matrix
response_matrix <- iris_df %>%
  dplyr::select(sepal_length, sepal_width, petal_length, petal_width) %>%
  as.matrix()

# Multivariate Shapiro-Wilk test
mshapiro.test(t(response_matrix))

```

Shapiro-Wilk normality test

data: Z
W = 0.97935, p-value = 0.02342

Interpretation: If $p < 0.05$, the assumption of multivariate normality is violated. MANOVA is fairly robust to moderate violations with large sample sizes.

Homogeneity of Covariance Matrices

```

# Prepare response variables
response_vars_df <- iris_df %>%
  dplyr::select(sepal_length, sepal_width, petal_length, petal_width)

```

```
# Box's M test for homogeneity of covariance matrices
iris_box_m_model <- boxM(response_vars_df, iris_df$species)
iris_box_m_model
```

Box's M-test for Homogeneity of Covariance Matrices

```
data: response_vars_df
Chi-Sq (approx.) = 140.94, df = 20, p-value < 2.2e-16
```

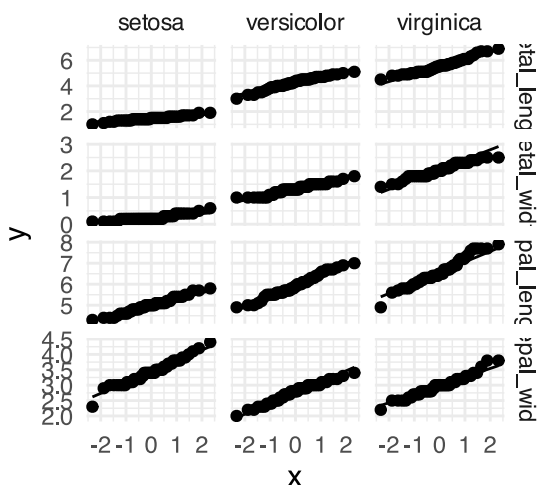
Interpretation: If $p < 0.05$, covariance matrices differ between groups. MANOVA is robust to this violation with equal sample sizes.

Visual Assessment of Normality

```
# Create Q-Q plots for each variable by species
iris_qq_plot <- iris_long_df %>%
  ggplot(aes(sample = measure)) +
  geom_qq() +
  geom_qq_line() +
  facet_grid(variable ~ species, scales = "free") +
  labs(title = "Q-Q Plots by Species and Variable") +
  theme_minimal()

iris_qq_plot
```

Q-Q Plots by Species and Variable



Step 3: Fit MANOVA Model

```
# Fit MANOVA model
iris_manova_model <- manova(cbind(sepal_length, sepal_width, petal_length, petal_width) ~
  species,
                           data = iris_df)

# View MANOVA results
summary(iris_manova_model)
```

```

          Df Pillai approx F num Df den Df    Pr(>F)
species    2 1.1919   53.466      8   290 < 2.2e-16 ***
Residuals 147
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Interpretation: - Pillai's trace is the default test statistic (most robust) - Large F-value and small p-value indicate significant group differences - The null hypothesis (all species have same multivariate means) is rejected

Step 4: Alternative Test Statistics

```

# Wilks' Lambda
summary(iris_manova_model, test = "Wilks")

```

```

          Df   Wilks approx F num Df den Df    Pr(>F)
species    2 0.023439   199.15      8   288 < 2.2e-16 ***
Residuals 147
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

# Hotelling-Lawley Trace
summary(iris_manova_model, test = "Hotelling-Lawley")

```

```

          Df Hotelling-Lawley approx F num Df den Df    Pr(>F)
species    2          32.477   580.53      8   286 < 2.2e-16 ***
Residuals 147
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

# Roy's Largest Root
summary(iris_manova_model, test = "Roy")

```

```

          Df   Roy approx F num Df den Df    Pr(>F)
species    2 32.192   1167      4   145 < 2.2e-16 ***
Residuals 147
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Step 5: Follow-up Univariate ANOVAs

Since MANOVA is significant, we examine which variables contribute to group differences.

```

# Extract univariate ANOVA results
summary.aov(iris_manova_model)

```

```

Response sepal_length :
          Df Sum Sq Mean Sq F value    Pr(>F)
species    2  63.212   31.606  119.26 < 2.2e-16 ***
Residuals 147  38.956    0.265
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

Response sepal_width :
      Df Sum Sq Mean Sq F value    Pr(>F)
species    2 11.345   5.6725   49.16 < 2.2e-16 ***
Residuals 147 16.962   0.1154
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Response petal_length :
      Df Sum Sq Mean Sq F value    Pr(>F)
species    2 437.10 218.551 1180.2 < 2.2e-16 ***
Residuals 147  27.22   0.185
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Response petal_width :
      Df Sum Sq Mean Sq F value    Pr(>F)
species    2 80.413  40.207  960.01 < 2.2e-16 ***
Residuals 147  6.157   0.042
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Step 6: Post-hoc Comparisons

```

# Sepal Length ANOVA and comparisons
sepal_length_model <- aov(sepal_length ~ species, data = iris_df)
summary(sepal_length_model)

```

```

      Df Sum Sq Mean Sq F value Pr(>F)
species    2  63.21  31.606  119.3 <2e-16 ***
Residuals 147  38.96   0.265
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

# Pairwise comparisons
sepal_length_emm <- emmeans(sepal_length_model, ~ species)
pairs(sepal_length_emm)

```

```

contrast      estimate      SE df t.ratio p.value
setosa - versicolor    -0.930 0.103 147  -9.033  <.0001
setosa - virginica     -1.582 0.103 147 -15.366  <.0001
versicolor - virginica -0.652 0.103 147  -6.333  <.0001

```

P value adjustment: tukey method for comparing a family of 3 estimates

```

# Sepal Width ANOVA and comparisons
sepal_width_model <- aov(sepal_width ~ species, data = iris_df)
summary(sepal_width_model)

```

```

      Df Sum Sq Mean Sq F value Pr(>F)
species    2  11.35   5.672   49.16 <2e-16 ***
Residuals 147  16.96   0.115

```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Pairwise comparisons
sepal_width_emm <- emmeans(sepal_width_model, ~ species)
pairs(sepal_width_emm)
```

contrast	estimate	SE	df	t.ratio	p.value
setosa - versicolor	0.658	0.0679	147	9.685	<.0001
setosa - virginica	0.454	0.0679	147	6.683	<.0001
versicolor - virginica	-0.204	0.0679	147	-3.003	0.0088

P value adjustment: tukey method for comparing a family of 3 estimates

```
# Petal Length ANOVA and comparisons
petal_length_model <- aov(petal_length ~ species, data = iris_df)
summary(petal_length_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
species	2	437.1	218.55	1180	<2e-16 ***
Residuals	147	27.2	0.19		

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Pairwise comparisons
petal_length_emm <- emmeans(petal_length_model, ~ species)
pairs(petal_length_emm)
```

contrast	estimate	SE	df	t.ratio	p.value
setosa - versicolor	-2.80	0.0861	147	-32.510	<.0001
setosa - virginica	-4.09	0.0861	147	-47.521	<.0001
versicolor - virginica	-1.29	0.0861	147	-15.012	<.0001

P value adjustment: tukey method for comparing a family of 3 estimates

```
# Petal Width ANOVA and comparisons
petal_width_model <- aov(petal_width ~ species, data = iris_df)
summary(petal_width_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
species	2	80.41	40.21	960	<2e-16 ***
Residuals	147	6.16	0.04		

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Pairwise comparisons
petal_width_emm <- emmeans(petal_width_model, ~ species)
pairs(petal_width_emm)
```

contrast	estimate	SE	df	t.ratio	p.value
setosa - versicolor	-1.08	0.0409	147	-26.387	<.0001
setosa - virginica	-1.78	0.0409	147	-43.489	<.0001
versicolor - virginica	-0.70	0.0409	147	-17.102	<.0001

P value adjustment: tukey method for comparing a family of 3 estimates

Step 7: Canonical Discriminant Analysis

To understand how the groups differ in multivariate space, we perform canonical discriminant analysis.

```
# Perform canonical discriminant analysis
iris_candisc_model <- candisc(iris_manova_model)
iris_candisc_model
```

Canonical Discriminant Analysis for species:

	CanRsq	Eigenvalue	Difference	Percent	Cumulative
1	0.96987	32.19193	31.907	99.12126	99.121
2	0.22203	0.28539	31.907	0.87874	100.000

Test of H0: The canonical correlations in the current row and all that follow are zero

	LR test stat	approx F	numDF	denDF	Pr(> F)
1	0.02344	199.145	8	288	< 2.2e-16 ***
2	0.77797	13.794	3	145	5.794e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# Eigenvalues
iris_candisc_model$eigenvalues
```

```
[1] 3.219193e+01 2.853910e-01 -7.801056e-17 -1.398429e-15
```

```
# Proportion of variance explained
prop_variance <- iris_candisc_model$eigenvalues / sum(iris_candisc_model$eigenvalues)
prop_variance
```

```
[1] 9.912126e-01 8.787395e-03 -2.402001e-18 -4.305862e-17
```

```
# Cumulative proportion
cumsum(prop_variance)
```

```
[1] 0.9912126 1.0000000 1.0000000 1.0000000
```

Step 8: Linear Discriminant Analysis

```
# Perform LDA
iris_lda_model <- lda(species ~ sepal_length + sepal_width + petal_length + petal_width,
                      data = iris_df)

iris_lda_model
```

```
Call:
lda(species ~ sepal_length + sepal_width + petal_length + petal_width,
    data = iris_df)
```

```
Prior probabilities of groups:
      setosa versicolor virginica
0.3333333  0.3333333  0.3333333
```

```
Group means:
      sepal_length sepal_width petal_length petal_width
setosa           5.006       3.428       1.462       0.246
versicolor       5.936       2.770       4.260       1.326
virginica         6.588       2.974       5.552       2.026
```

```
Coefficients of linear discriminants:
      LD1      LD2
sepal_length  0.8293776 -0.02410215
sepal_width   1.5344731 -2.16452123
petal_length -2.2012117  0.93192121
petal_width  -2.8104603 -2.83918785
```

```
Proportion of trace:
      LD1      LD2
0.9912  0.0088
```

```
# Get predictions
iris_lda_pred <- predict(iris_lda_model)

# Create dataframe with LDA scores
lda_scores_df <- data.frame(
  LD1 = iris_lda_pred$x[, 1],
  LD2 = iris_lda_pred$x[, 2],
  species = iris_df$species
)
```

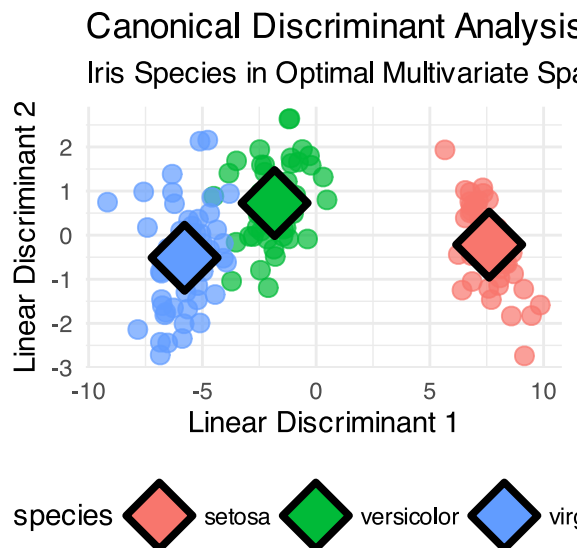
Step 9: Visualize Canonical Space

```
# Calculate group centroids in canonical space
centroids_df <- lda_scores_df %>%
  group_by(species) %>%
  summarise(
    LD1_mean = mean(LD1),
    LD2_mean = mean(LD2),
    .groups = 'drop'
  )

# Create canonical plot
canonical_plot <- lda_scores_df %>%
  ggplot(aes(x = LD1, y = LD2, color = species)) +
```

```
geom_point(size = 3, alpha = 0.7) +
geom_point(data = centroids_df,
  aes(x = LD1_mean, y = LD2_mean, fill = species),
  shape = 23, color = "black", size = 8, stroke = 2) +
labs(title = "Canonical Discriminant Analysis",
  subtitle = "Iris Species in Optimal Multivariate Space",
  x = "Linear Discriminant 1",
  y = "Linear Discriminant 2") +
theme_minimal() +
theme(legend.position = "bottom")
```

canonical_plot



Step 10: Effect Size

```
# Calculate Wilks' Lambda for effect size
manova_wilks <- summary(iris_manova_model, test = "Wilks")
wilks_lambda <- manova_wilks$stats[1, "Wilks"]
wilks_lambda
```

```
[1] 0.02343863
```

```
# Calculate partial eta-squared
partial_eta_squared <- 1 - wilks_lambda
partial_eta_squared
```

```
[1] 0.9765614
```

Summary Checklist for MANOVA

When conducting MANOVA, always follow these steps:

💡 MANOVA Checklist

1. **Visualize your data** - boxplots and scatterplots by groups
2. **Check assumptions**
 - Multivariate normality (Shapiro-Wilk test)
 - Homogeneity of covariance matrices (Box's M test)
 - Visual assessment with Q-Q plots
3. **Fit MANOVA model** - response variables ~ grouping factor
4. **Examine test statistics** - Pillai's, Wilks', Hotelling-Lawley, Roy's
5. **Follow-up analyses** if MANOVA is significant
 - Univariate ANOVAs for each variable
 - Post-hoc pairwise comparisons
6. **Canonical analysis** - understand multivariate patterns
7. **Calculate effect size** - partial eta-squared from Wilks' Lambda
8. **Visualize results** - canonical plots showing group separation

Key Points to Remember

- **MANOVA controls Type I error** when testing multiple dependent variables
- **More powerful than separate ANOVAs** when variables are correlated
- **Tests group centroids** in multivariate space, not individual means
- **Canonical variates** show optimal linear combinations for group separation
- **Effect sizes** can be very large when groups are well-separated
- **Assumptions are more stringent** than univariate ANOVA

! Key Points from MANOVA Analysis

1. **Check multivariate assumptions first** - normality and homogeneity of covariances
2. **MANOVA tests the global hypothesis** - do groups differ on any combination of variables?
3. **Follow-up tests identify specific differences** - which variables drive group separation
4. **Canonical analysis reveals patterns** - how variables work together to discriminate groups
5. **Visualize in reduced space** - canonical plots show multivariate relationships clearly
6. **Interpret effect sizes** - Wilks' Lambda tells us proportion of variance explained
7. **Consider biological meaning** - what do the multivariate patterns tell us about the organisms?

Remember: MANOVA is ideal when you expect groups to differ on multiple correlated traits that reflect an integrated biological system!